



# Search-Skills-Writeup

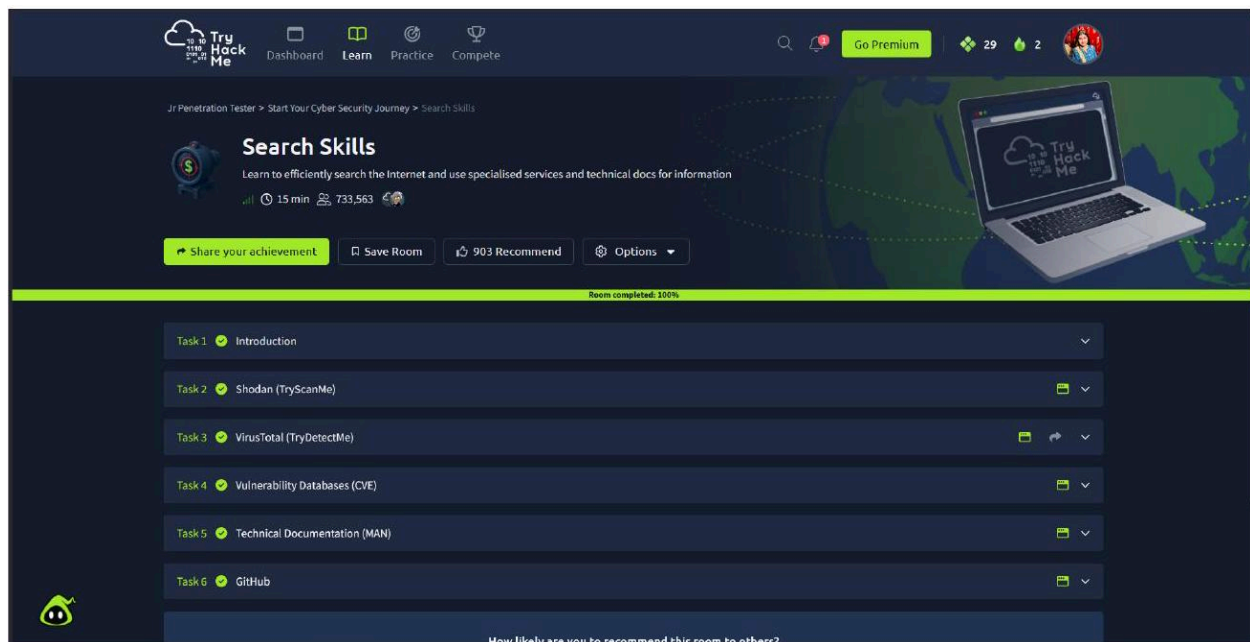
## Search Skills

TryHackMe Write-up

<https://tryhackme.com/room/searchskills>

## What This Room Is About

This room is about learning to search the internet the smart way - using the right tool for the right job instead of just plain Googling. It covers five tools: Shodan, VirusTotal, CVE databases, Linux man pages, and GitHub.



Room completed - 100%

## 1. Shodan

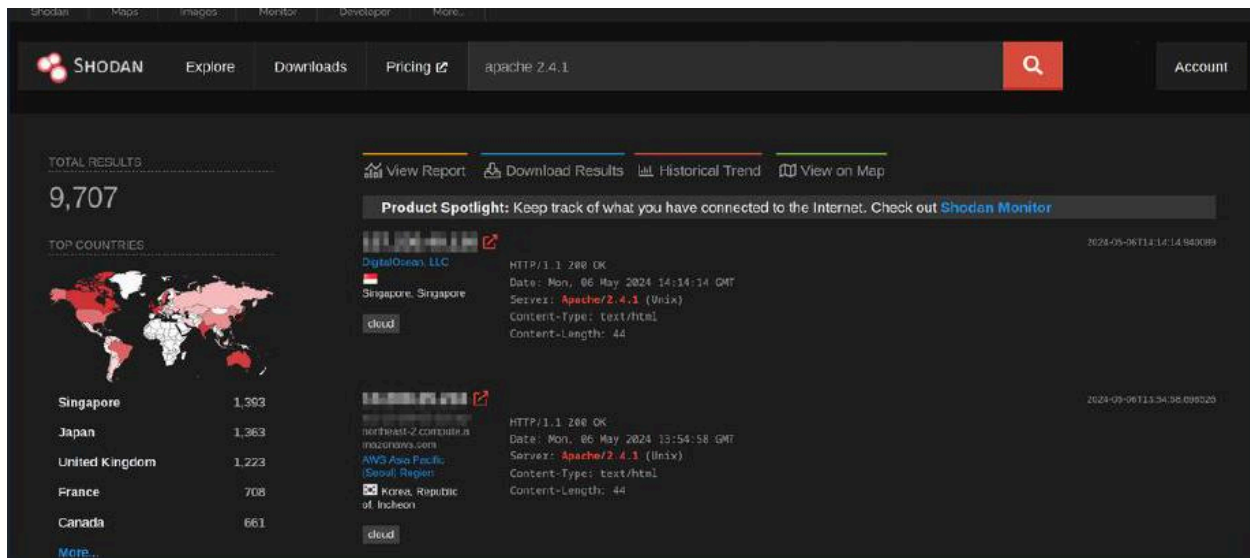
What it does

Shodan is like Google, but instead of websites, it searches for devices connected to the internet - servers, cameras, routers, anything with an open connection.

Steps

- Went to Shodan and searched for a software name and version, like apache 2.4.1
- Got back a list of servers running that exact version, with their location and company
- Used filters like country:, port:, org:, and hostname: to narrow the search down

Shodan is often described as a search engine for the Internet of Things (IoT), but that undersells it. Shodan continuously scans the internet, searching for networking equipment, industrial control systems, traffic cameras, and virtually anything else with a public network connection to see what's running and where.



For example, searching apache 2.4.1 will return a list of servers advertising that version in their HTTP headers, broken down by country, organisation, and port. During a penetration test or vulnerability assessment, that kind of visibility is extremely useful, particularly when paired with a known CVE affecting that version.

Shodan also supports its own query filters, which let you narrow results significantly:

| Filter  | Description                                  | Example                              |
|---------|----------------------------------------------|--------------------------------------|
| country | Restrict results to a specific country code. | <input type="checkbox"/> country: IE |

## Searching Shodan for a specific server version

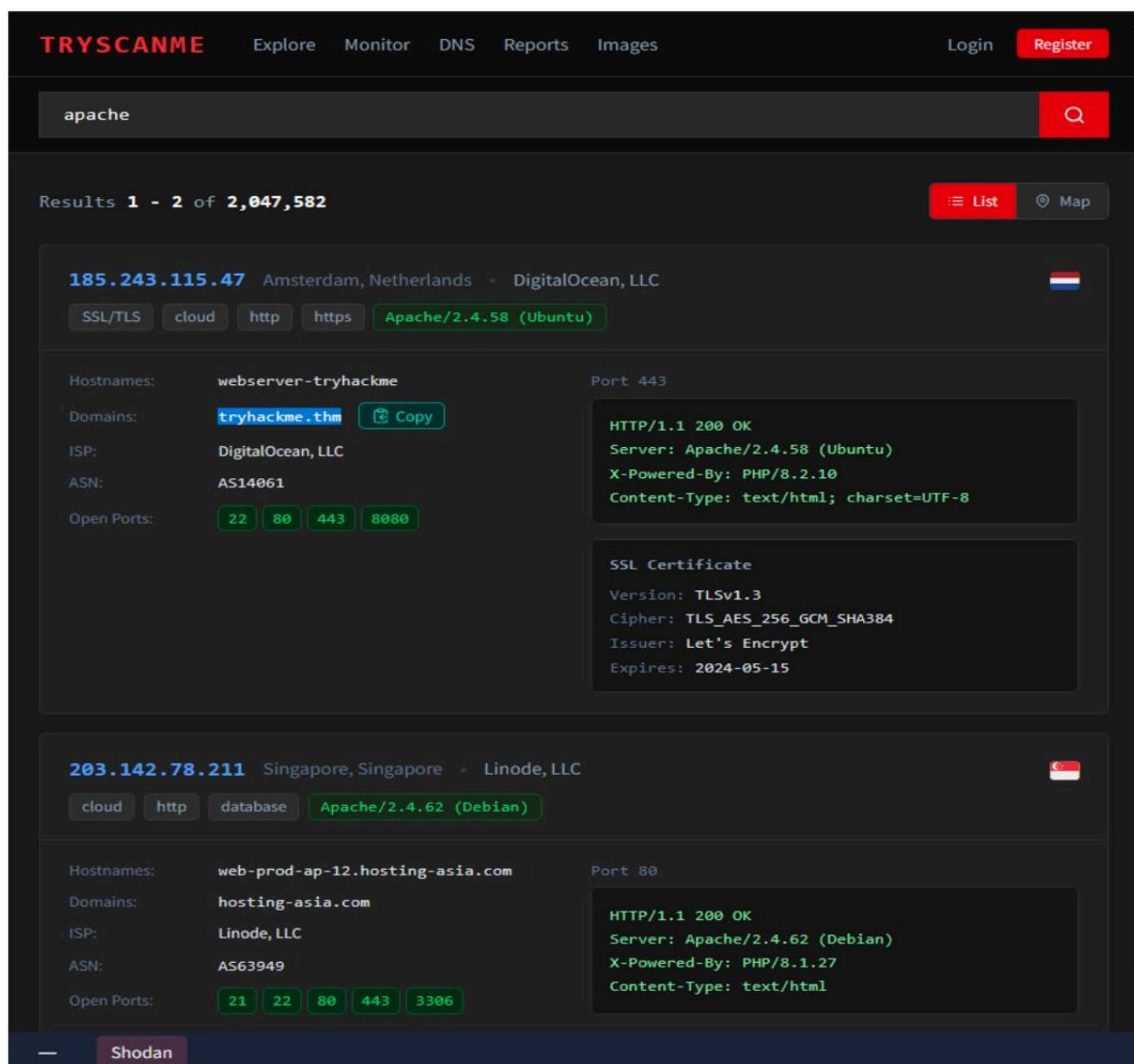
Shodan also supports its own query filters, which let you narrow results significantly:

| Filter   | Description                                                                                 | Example                                        |
|----------|---------------------------------------------------------------------------------------------|------------------------------------------------|
| country  | Restrict results to a specific country code.                                                | <input type="checkbox"/> country: IE           |
| port     | Filter by a specific port number or a range.                                                | <input type="checkbox"/> port: 22              |
| org      | Scope results in a named organisation or ASN Identifier (Who owns a range of IP addresses). | <input type="checkbox"/> (Amazon Web Services) |
| hostname | Match against a specific hostname or domain.                                                | <input type="checkbox"/> hostname:fakebank.thm |

## Practical

Click the view site button below to start our Shodan simulation:

[View Site](#)



Practice site - real-looking results for an Apache search

Real life use: Before attacking or defending a system, this is how you check what's actually exposed to the internet, and whether it's running an old, unsafe version of something.

## 2. VirusTotal

What it does

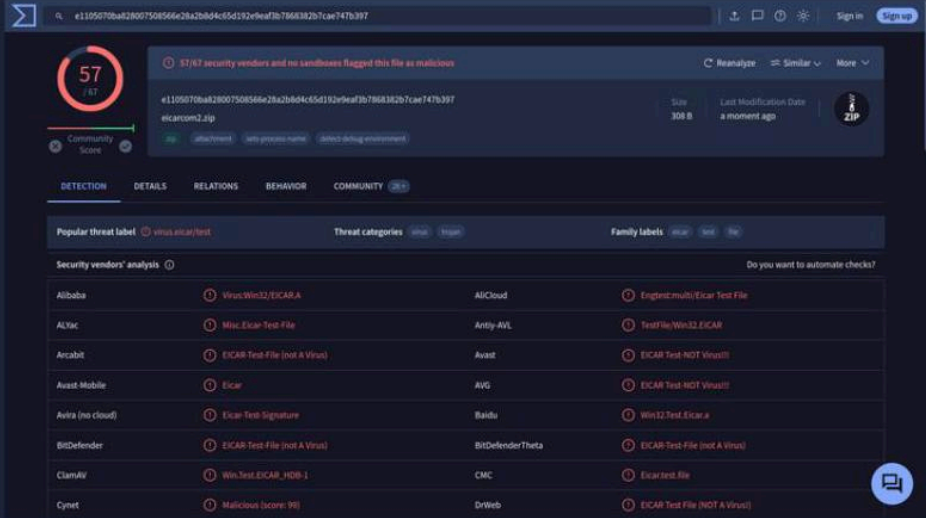
VirusTotal checks a file, link, or website against over 70 antivirus engines at once, and tells you if any of them think it's dangerous.

## Steps

- Searched for a file (invoice\_payment.exe) on a VirusTotal-style site
- Saw a score showing how many security vendors flagged it as malicious
- Checked the vendor list to see what type of threat they called it

**Task 3** VirusTotal (TryDetectMe)

VirusTotal collates results from over 70 antivirus engines and website scanners into a single interface. Submit a file, a URL, a domain, or a file hash. VirusTotal will tell you whether any of those engines have flagged it as malicious or not.



**Community Score** 57/87

37/87 security vendors and sandboxes flagged this file as malicious

File: e1195670ba12800750856a28a2b8d4c65d192e9eaf7b786838297cae747b397  
eicarcom2.zip

Size: 308 B | Last Modification Date: a moment ago

**DETECTION** DETAILS RELATIONS BEHAVIOR COMMUNITY

Popular threat label: virus:win32/eicar Test Threat categories: Similar Family labels: eicar Test File

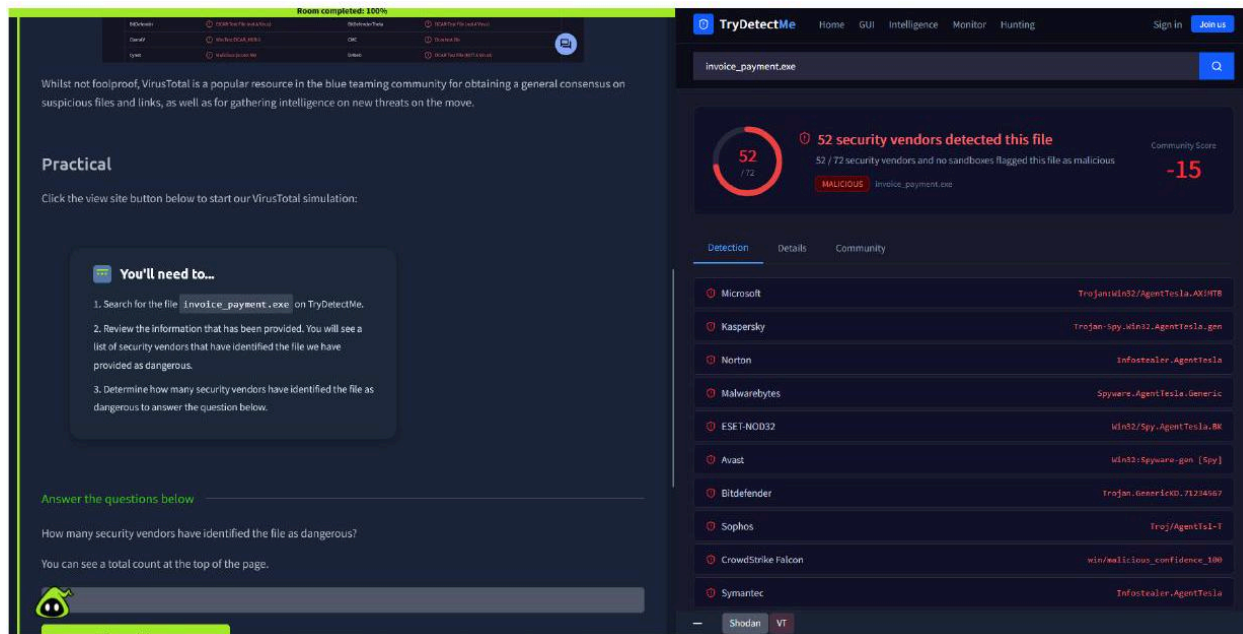
Security vendors' analysis

| Vendor                          | Threat Label                  |
|---------------------------------|-------------------------------|
| Alibaba                         | Virus:Win32/EICAR.A           |
| ALYac                           | Misc.Eicar-Test-File          |
| Arcabit                         | EICAR-Test-File (not A Virus) |
| Avast Mobile                    | Eicar                         |
| Avira (no cloud)                | Eicar-Test-Signature          |
| BitDefender                     | EICAR-Test-File (not A Virus) |
| ClamAV                          | Win.Test.EICAR_HDB-1          |
| Cynet                           | Malicious (Score: 98)         |
| Engtest-multi/Eicar-Test-File   |                               |
| TestFile/Win32.EICAR            |                               |
| EICAR-Test-NOT-Virus!!          |                               |
| EICAR-Test-NOT-Virus!!          |                               |
| Win32.Test.Eicar.a              |                               |
| EICAR-Test-File (not A Virus)   |                               |
| Eicar-test-File                 |                               |
| EICAR-Test-File (NOT A Virus!!) |                               |

Whilst not foolproof, VirusTotal is a popular resource in the blue teaming community for obtaining a general consensus on suspicious files and links, as well as for gathering intelligence on new threats on the move.



## How VirusTotal shows its results



Practice site - a file flagged as malicious by 52 out of 72 vendors

Real life use: If someone sends you a suspicious file or link, this is the quick way to check if it's actually dangerous before opening it.

### 3. CVE Databases

What it does

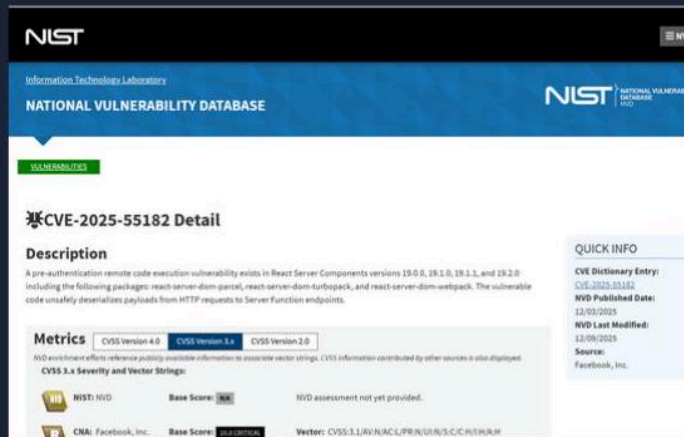
CVE stands for Common Vulnerabilities and Exposures. It's a public list of known security bugs, each with its own ID number and a score showing how serious it is.

Steps

- Searched a CVE ID (like CVE-2026-1337) in a vulnerability database
- Read the description to see what the bug actually does
- Checked the CVSS score to see how severe it is - this one scored 10 out of 10, the highest possible

#### Task 4 Vulnerability Databases (CVE)

The **Common Vulnerabilities and Exposures (CVE)** programme is the closest thing the industry has to a universal dictionary of known vulnerabilities.



Each confirmed vulnerability is assigned a unique identifier in the format **CVE-YEAR-NUMBER**, such as **CVE-2025-55182**. If the vulnerability is impactful enough, it may even get a moniker. You may have heard of vulnerabilities such as Heartbleed, React2Shell, and Log4Shell. These vulnerabilities are given a score (**CVSS**) based on a variety of factors, such as:

- **Impact** - What damage can this vulnerability lead to?
- **Complexity** - Is the vulnerability easy to exploit or not?
- **Availability** - How likely is it that someone can exploit this?

Organisations use scoring like this to prioritise their level of risk. Addressing the highest scoring first.

These identifiers function as a reference point among vendors, researchers, security tools, and documentation, ensuring that anyone discussing a vulnerability refers to the same issue. Websites like ExploitDB compile this information alongside "Proof of Concepts" (PoCs), which are scripts capable of demonstrating the vulnerability.

A CVE entry, with its score and details

Room completed: 100%

These identifiers function as a reference point among vendors, researchers, security tools, and documentation, ensuring that everyone discussing a vulnerability refers to the same issue. Websites like ExploitDB compile this information alongside "Proof of Concepts" (PoCs), which are scripts capable of demonstrating the vulnerability.

| Date     | ID | A | V | Title                                              | Type   | Platform | Author           |
|----------|----|---|---|----------------------------------------------------|--------|----------|------------------|
| 20200508 | ✓  | ✓ | ✓ | Apache Commons 4.7.0 (CVE-2020-1337) SQL Injection | WebApp | Multiple | Exploit-Research |
| 20200508 | ✓  | ✓ | ✓ | Apache HBase 4.11.0 Remote Code Execution (RCE)    | Remote | Multiple | Exploit-Research |
| 20200509 | ✓  | ✓ | ✓ | Drupal 9.3.2 (0-Remote Code Execution)             | WebApp | Python   | Exploit-Research |
| 20200509 | ✓  | ✓ | ✓ | Drupal 9.3.2 (0-Remote Code Execution)             | WebApp | Python   | Exploit-Research |
| 20200509 | ✓  | ✓ | ✓ | Drupal 9.3.2 (0-Remote Code Execution)             | WebApp | Python   | Exploit-Research |
| 20200509 | ✓  | ✓ | ✓ | Drupal 9.3.2 (0-Remote Code Execution)             | WebApp | Python   | Exploit-Research |

**Practical**

For this section, you will be interacting with TryHackMe's Vulnerability Database. Click the "Show Site" button below to get started.

**You'll need to...**

1. Search the Vulnerability Database for **CVE-2026-1337**
2. Review the details about the vulnerability
3. Find the CVSS scoring to answer the question.

**Answer the questions below**

What **CVSS** (Common Vulnerability Scoring System) classification did the vulnerability get?

10

**Vulnerability Database**

Vulnerabilities Products Metrics Configurations About

CVE Search / Vulnerability Detail

**CVE-2026-1337**

**CVE-2026-1337** **CRITICAL**

Published: January 22, 2026 Last Modified: February 15, 2026

**DESCRIPTION**

A critical SQL injection vulnerability exists in Apache WebPortal versions 3.2.0 through 3.4.5 that allows remote attackers to execute arbitrary SQL commands via the "userid" parameter in the login authentication module. This vulnerability can be exploited without authentication and may lead to complete database compromise, including unauthorized access to sensitive user credentials, session tokens, and confidential application data.

**CVSS SCORE**

**10** /10 **HIGH** CVSS Version 2.0  
AV:N/AC:L/Au:N/C:C/I:C/A:C

**CVSS METRICS**

| Attack Vector    | Attack Complexity | Link                   |
|------------------|-------------------|------------------------|
| Authentication   | NONE              | CONFIDENTIALITY IMPACT |
| Integrity Impact | COMPLETE          | AVAILABILITY IMPACT    |

**WEAKNESS ENUMERATION**

**CVE-89 SQL Injection**

**VULNERABILITY DETAILS**

Practice site - CVE-2026-1337, a critical SQL injection bug

Real life use: Before fixing or patching anything, this is how you find out exactly what a bug does and how urgent it is to fix.

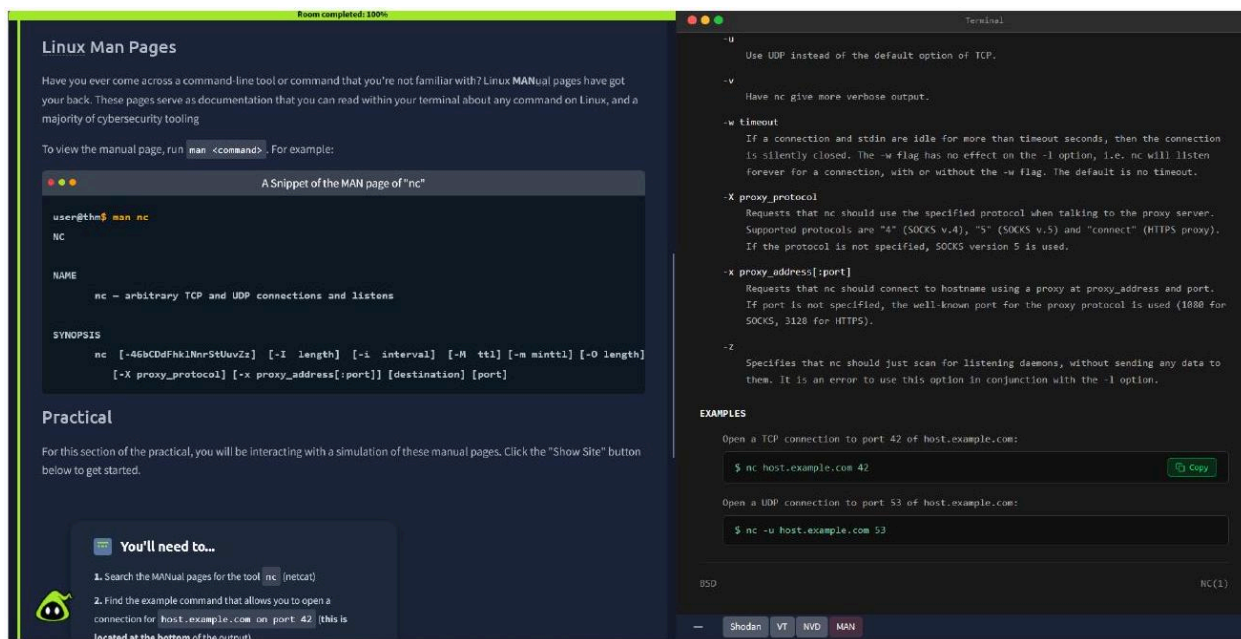
## 4. Linux Man Pages

What it does

Man pages are the built-in instruction manual for almost every command on Linux. If you don't know how a command works, its man page will tell you.

Steps

- Ran man nc in the terminal to open the manual for the netcat tool
- Read through the options and flags it supports
- Found an example command in the manual and copied its usage



The man page for nc, with example commands

Real life use: Whenever you're stuck on how to use a command-line tool, this is the fastest and most reliable place to check - no need to search the web.

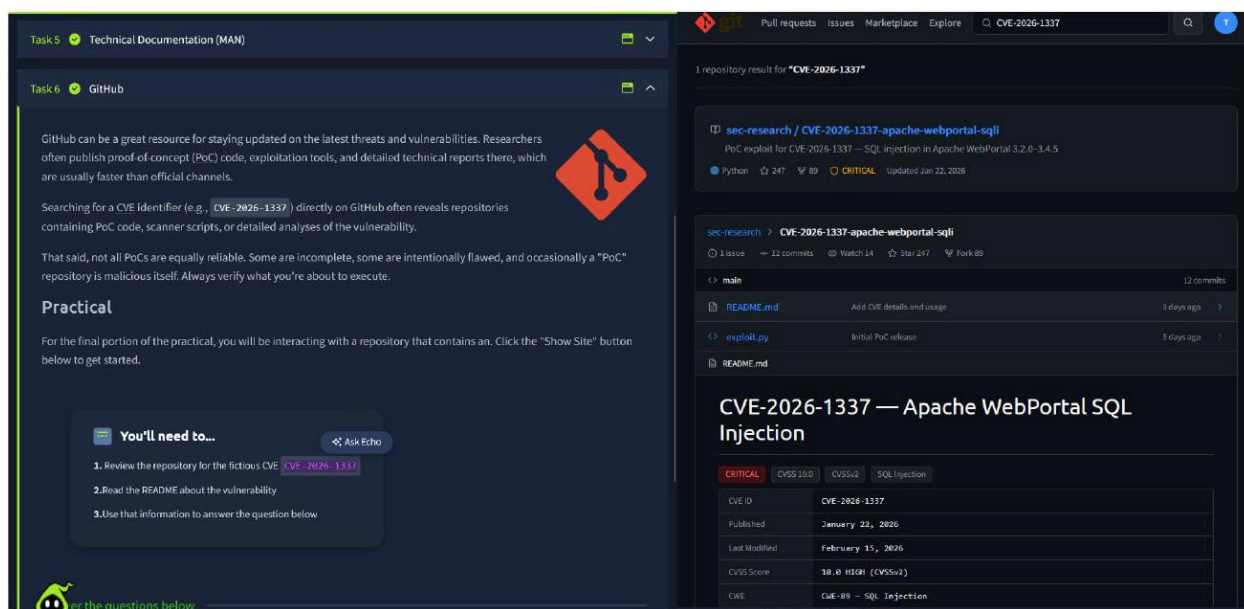
## 5. GitHub

What it does

GitHub is where a lot of security researchers share their write-ups, tools, and proof-of-concept code for vulnerabilities they've found.

Steps

- Searched a CVE ID directly on GitHub
- Found a repository someone had made about that exact bug
- Read the README to understand the bug and see the proof-of-concept code



Real life use: This is how you find real working examples of a bug, and how attackers or defenders alike learn what a vulnerability looks like in practice. Worth noting: not every PoC on GitHub is trustworthy, so always read the code before you run it.

## Wrap Up

None of these tools need hacking skills on their own - they're search skills. Shodan finds exposed devices, VirusTotal checks if a file is safe, CVE databases explain a bug, man pages explain a command, and GitHub shows real examples. Together, they're the first thing most security people reach for before doing anything else.

[^0]

[^0]: Write-up by FireWallFairy - TryHackMe, Jr Penetration Tester path.